AGENTS.md: instructions for AI coding agents working on Typographics Art This file tells AI agents (and humans) how to work in this repository. Read it before changing anything. ## The project in one paragraph Typographics Art sets a given text, character for character and in order, so that only the typography of each glyph (size, weight, colour; width is an optional axis, off by default) changes and the page forms a picture. The text must stay real, extractable text. The three rules in `docs/technique.md` are the contract of the project; no change may break them. ## Layout - `src/`: the library. Pure modules (`layout.js`, `tone.js`, `color.js`, `render.js`, `typeset.js`, `fonts.js`, `options.js`, `lorem.js`) run in Node and browsers. `browser.js` needs a DOM: font measurement, image reading, PNG output. - `index.html`: the landing page. `app.html`: the generator (served at `/app.html`). `work.html`: the three views of a work (Art, Read, Machine). `gallery.html`, `examples.html`, `404.html`: gallery, one-pager examples, error page. - `cli/`: command-line tools that drive headless Chrome over the DevTools protocol (`core.mjs`, `render.html`), the renderer `tg-art.mjs`, the one-pager `onepage.mjs`, `tg-secret.mjs` for hidden messages, gallery and asset builders. - `api/`: the paid one-pager API server (see `api/README.md`). - `scripts/`: builders for the website, the samples, `about/`, and the measurement scripts behind `docs/`. - `gallery/`: the 15 works, their photos and `works.config.json`. - `examples/`: one-pagers (31) of laws, declarations and AI instructions with their published source texts. - `samples/`: sample texts for the generator, generated by `scripts/build-samples.mjs`. - `demo/moby-dick/`: the demonstrator (whole book as a poster) with its proof report. - `docs/`: measurements, the definition of the technique and the rules of the project. - `about/`: the generated single-file description for AI assistants (do not edit by hand; run `node scripts/build-about.mjs`). - `benchmark/`: the test document for `docs/ai-benefit.md`. - `assets/`: brand assets. - `test/`: Node's built-in test runner, no dependencies. ## Rules 1. **No runtime dependencies and no build step for the library.** Plain ES modules only. The website needs `node scripts/build-site.mjs`, which is only used to deploy it. Pages load no third-party resources; vendored files are listed in `NOTICE.md`. 2. **Fidelity is tested, not assumed.** Any change to layout, rendering or export must keep `isFaithful()` true and the text extractable from HTML, SVG and PDF. Add or update a test in `test/` for every behaviour change. 3. **Pure logic stays pure.** New algorithms go into the pure modules with tests that run under `node --test`. Only code that needs a DOM goes into `browser.js`, `cli/render.html` or the pages. 4. **Measure before claiming.** Numbers in the documentation (fidelity, OCR rates, timings) must come from a script or test in this repository. Update them when the code changes. 5. **Keep outputs honest.** Generated reports must state their method and limits. Never hide instructions in tiny text: a one-pager always shows its title, and its full text must be the declared document. 6. **English** for code, comments, commits and documentation; German translations live in `README.de.md` and the app's dictionary. ## Commands ```sh npm test # all tests, a few seconds npm run dev # needs Python 3; landing page at http://localhost:8080, generator at /app.html node cli/tg-art.mjs img.jpg --text t.txt -o out.pdf node cli/onepage.mjs doc.pdf --title "Title" -o onepage.pdf npm run gallery # re-render gallery (needs Chrome) node examples/build.mjs # re-render examples (needs Chrome and MuPDF; qpdf optional, shrinks the PDFs) node scripts/build-about.mjs # regenerate about/typographics-art.md (and its PDF; needs Chrome and MuPDF) ``` ## Tools - Node: library >= 20; CLI >= 22 (it needs the global `WebSocket`); API >= 22.5 (`node:sqlite`). - Chromium or Chrome (`CHROME_PATH` if it is not found), MuPDF (`mutool`), qpdf. - For specific scripts: pandoc and tesseract (`scripts/ai-benefit.mjs`); Python 3 with pypdf, pymupdf and pdfminer.six (`scripts/extractor-check.py`); Python 3 for `npm run dev`. ## Before you open a pull request - `npm test` passes. - For visual changes: render at least one gallery work or example and look at it. - The changelog has an entry if users will notice the change. - The pull request says what changed and how you checked it.